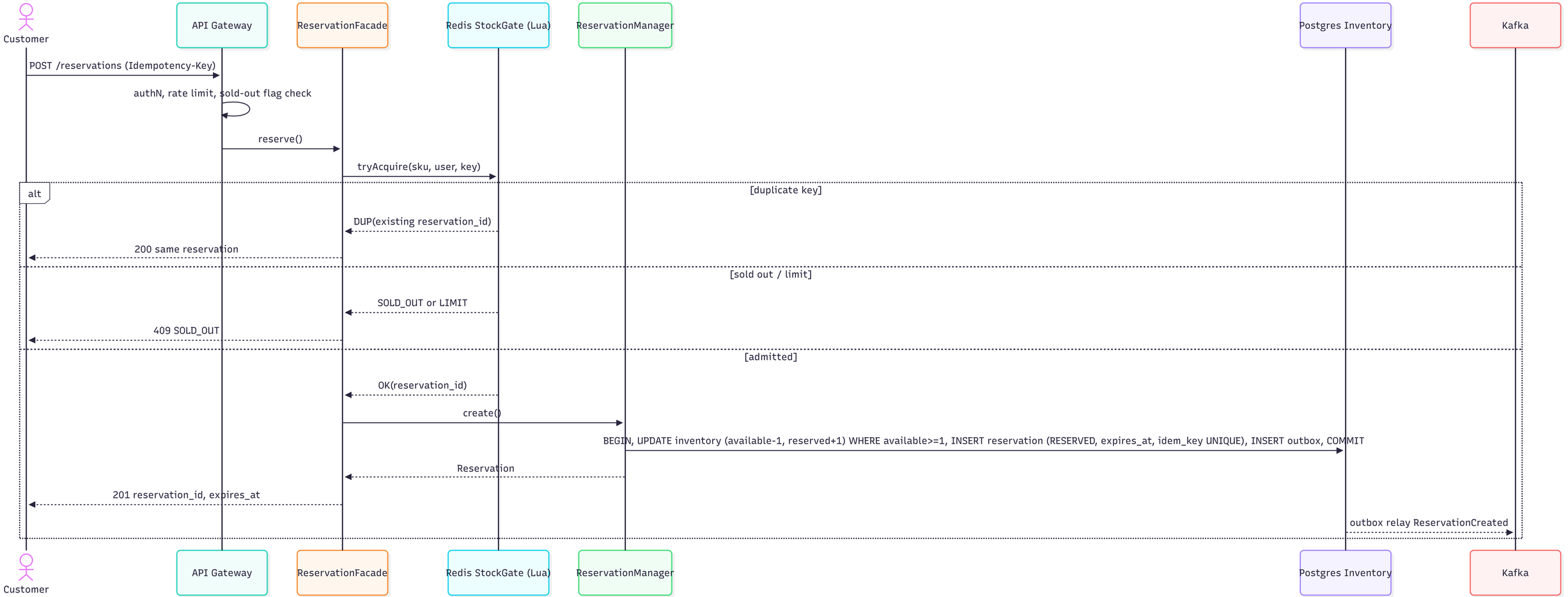

Customer
API Gateway
ReservationFacade
Redis StockGate (Lua)
ReservationManager
Postgres Inventory
Kafka
POST /reservations (Idempotency-Key)
authN, rate limit, sold-out flag check
reserve()
tryAcquire(sku, user, key)
alt
[duplicate key]
DUP(existing reservation_id)
200 same reservation
[sold out / limit]
SOLD_OUT or LIMIT
409 SOLD_OUT
[admitted]
OK(reservation_id)
create()
BEGIN, UPDATE inventory (available-1, reserved+1) WHERE available>=1, INSERT reservation (RESERVED, expires_at, idem_key UNIQUE), INSERT outbox, COMMIT
Reservation
201 reservation_id, expires_at
outbox relay ReservationCreated
Customer
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